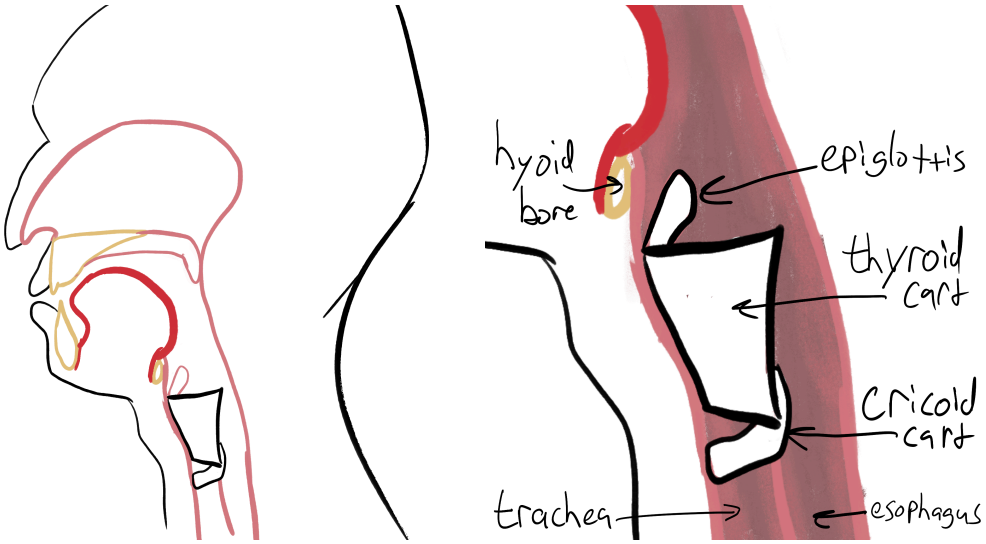
# Speech Anatomy 3

## The Larynx

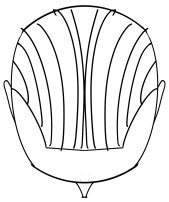
Dr. Josef Fruehwald

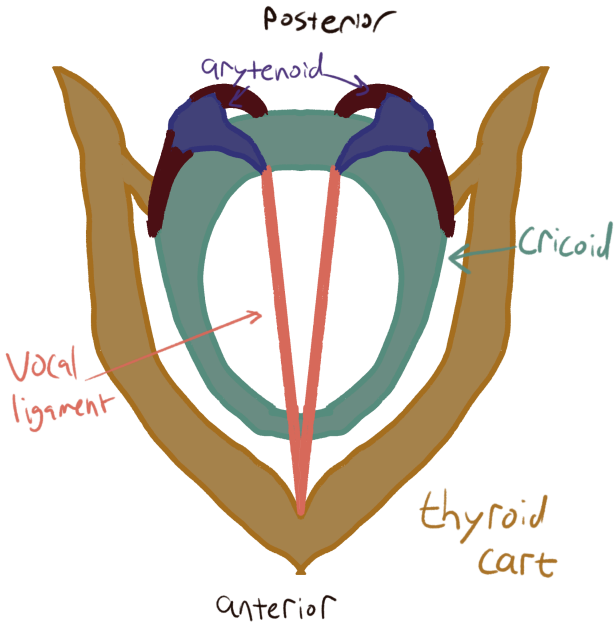
2026-09-01

## The Larynx

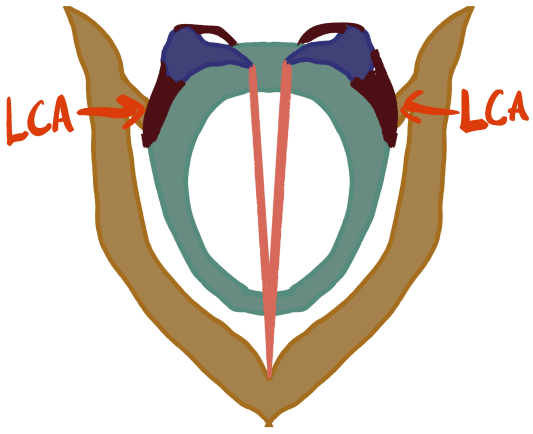


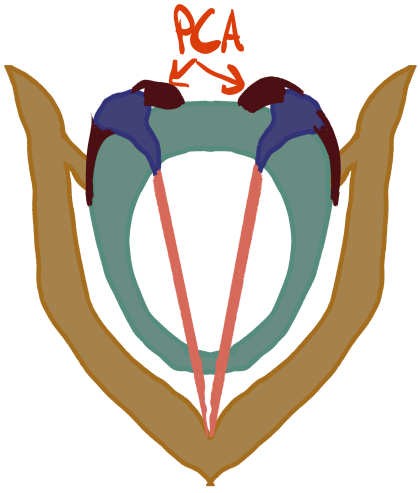
## Voicing






## Voicing

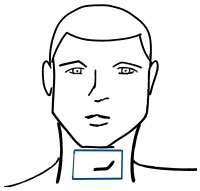


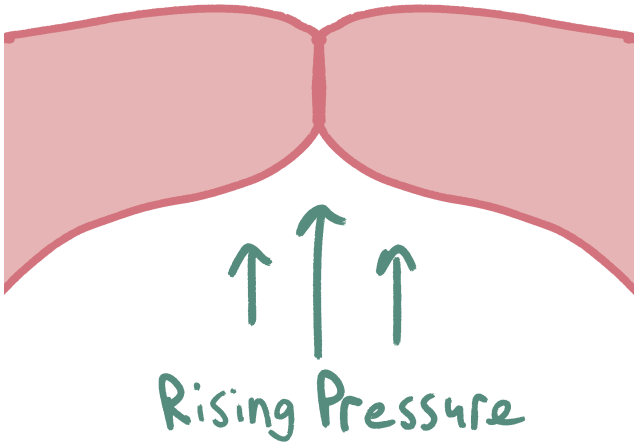




## Voicing

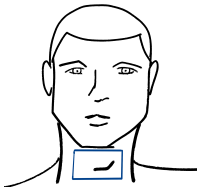
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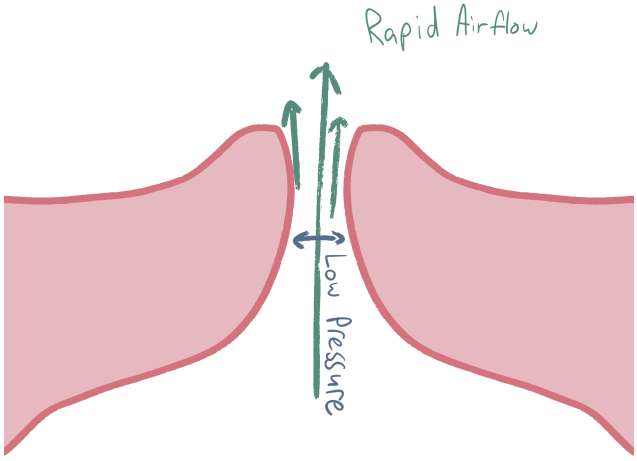
Voicing






Voicing





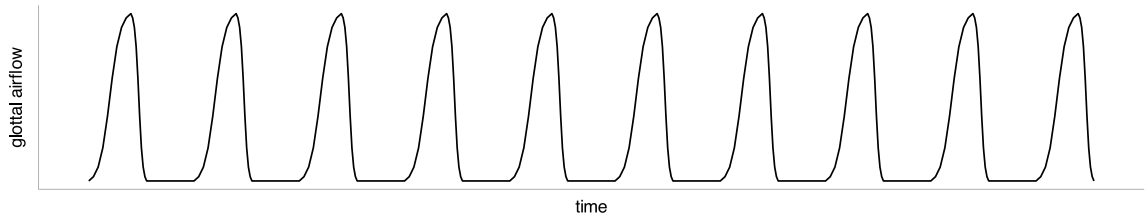
## Voicing

### Pitch

## Voice Quality

### Modal Voice

```
── Attaching core tidyverse packages ──────────────────────── tidyverse 2.0.0 ──
✓ dplyr     1.2.1     ✓ readr     2.2.0
✓ forcats   1.0.1     ✓ stringr   1.6.0
✓ ggplot2   4.0.3     ✓ tibble    3.3.1
✓ lubridate 1.9.5     ✓ tidyr     1.3.2
✓ purrr     1.2.2
── Conflicts ────────────────────────────────────────── tidyverse_conflicts() ──
✖ dplyr::filter() masks stats::filter()
✖ dplyr::lag()    masks stats::lag()
ℹ Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```



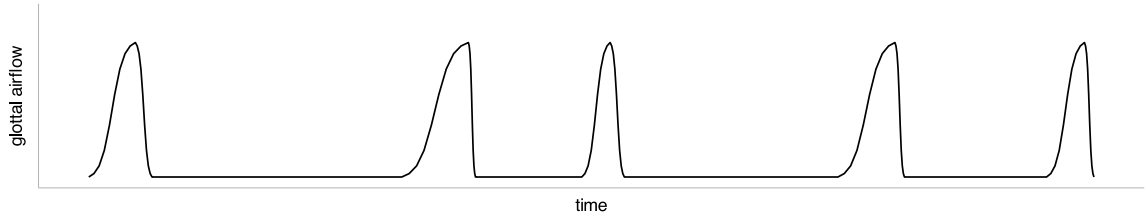

## Voice Quality

### Modal Voice

- Very “periodic”-glottal pulses at a regular rhythm
- High frequency

## Voice Quality

### Creaky Voice




## Voice Quality

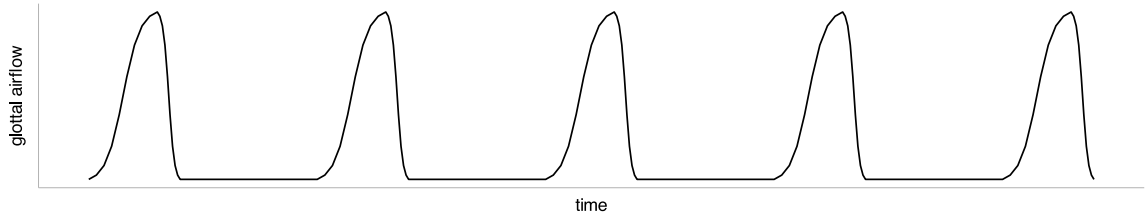
### Creaky Voice

- Low frequency
- Some irregular timing between opening phases
- Lower airflow
- Achieved by highly constricted glottis

Keating et al. (2015)

## Voice Quality

### Vocal Fry




## Voice Quality

### Vocal Fry

- Periodic (regular intervals)
- Low frequency
- Achieved by ??

Keating et al. (2015)

## Voice Quality

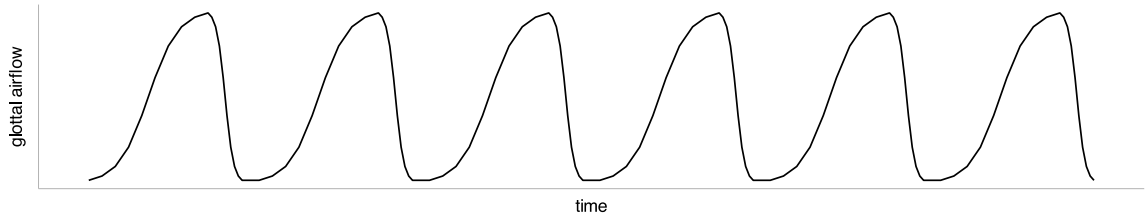
### Vocal Fry

Almost everything you've heard about "vocal fry" in popular culture is bunk.

- There's nothing harmful about it.
- People really only complain about women doing it, even though men also do it (Slobe, 2016)
- When people complain, they often also don't correctly identify what they're complaining about.

## Voice Quality

### Breathy voice




## Voice Quality

### Breathy Voice

- Periodic
- High glottal airflow
- Achieved by laxer vocal fold constriction

## References

## Bibliography

Keating, P. A., Garellek, M., & Kreiman, J. (2015). Acoustic properties of different kinds of creaky voice. *Proceedings of the International Congress of Phonetic Science XVIII*, (1). https://doi.org/10.1016/0167-8809(91)90080-H

Slobe, T. (2016,). This American Creak: Metaphors of Virus, Infection, and Contagion in Girls' Social Networks. *Paper Presented at AAA 2016*.